

Setup Guide

Overview

This guide provides instructions for setting up the greenfield_mp project locally, including environment configuration, database setup, and initial data loading.

Prerequisites

System Requirements

- Python 3.10+
- MySQL 8.0.19 or newer (or compatible like MariaDB 10.11)
- Git
- MySQL Workbench (recommended for database management)

Installation Steps

1. Clone the Repository

```
git clone https://github.com/abeer10-coder/greenfield_mp.git
cd greenfield_mp
```

2. Create Virtual Environment

```
python -m venv .venv
# Windows:
.venv\Scripts\activate
# macOS/Linux:
source .venv/bin/activate
```

3. Install Dependencies

```
pip install -r requirements.txt
```

4. Configure Environment

```
cp .env.example .env
# Edit .env with your MySQL credentials
# Required fields:
# MYSQL_HOST, MYSQL_PORT, MYSQL_USER, MYSQL_PASSWORD, MYSQL_DATABASE
```

5. Download Sample Data

```
# Option 1: Download IBM HR dataset from Kaggle
# Option 2: Use demo data generator
```

```
python scripts/synthesize_data.py --demo
```

Database Setup

Option 1: Using SQL Files (Recommended)

```
# Connect to MySQL
mysql -h YOUR_HOST -P YOUR_PORT -u YOUR_USER -p YOUR_DATABASE

# Run setup scripts in order:
mysql -h YOUR_HOST -P YOUR_PORT -u YOUR_USER -p greenfield_mp <
sql/01_schemas_and_staging.sql
mysql -h YOUR_HOST -P YOUR_PORT -u YOUR_USER -p greenfield_mp < sql/02_oltp_ddl.sql
mysql -h YOUR_HOST -P YOUR_PORT -u YOUR_USER -p greenfield_mp < sql/03_olap_ddl.sql
```

Option 2: Using Python Scripts

```
python scripts/setup_database.py
```

Data Loading

Step 1: Load Staging Data

```
python scripts/load_staging.py
```

Step 2: Load OLTP Data

```
python scripts/run_etl.py
# Or manually:
python scripts/setup_database.py
python scripts/load_staging.py
```

Step 3: Verify Data

```
python scripts/smoke_test.py
```

Streamlit App Setup

1. Configure Secrets

```
# Edit .streamlit/secrets.toml.example
# Set your MySQL connection details
```

2. Run the App

```
streamlit run app/streamlit_app.py
```

Troubleshooting

- **"Can't reach MySQL"**: Check .env file and ensure MySQL server is running
- **Access denied**: Verify credentials in .env file
- **Database not found**: Ensure database name matches in .env
- **SSL issues**: Ensure MySQL server supports SSL connections

Documentation

- Detailed architecture documentation: [docs/ARCHITECTURE.md](#)
- Git workflow: [docs/GIT_WORKFLOW.md](#)
- Database schema: [docs/diagrams/oltp_er_diagram.mmd](#) and [docs/diagrams/olap_star_schema.mmd](#)